

Music-Gen — _infra/harness-auto-write-namespacing (cycle 1, fork cc548ca0c2e5, clone 0)

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Contents

Music-Gen — _infra/harness-auto-write-namespacing (cycle 1, fork cc548ca0c2e5, clone 0)	1
Abstract	1
Introduction	2
Approach	2
Findings	3
Discussion	4
Open Questions	5
Appendix: Provenance	5

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Abstract

Cycle 1 of clone 0 retired cycle-21 handoff #1 by fixing the harness auto-write namespace collision at source rather than at reconciliation time. The upstream harness now namespaces `_run/report_cycles_<lo>-<hi>` per clone at write time (`_run/report_cycles_<lo>-<hi>_clone-<k>` when running under a fan-out clone context), so future 2+-clone fork merges concat cleanly through the standard `concat_clone_ledgers` path without the per-clone id normalization the cycle-22 driver had to apply. The SSoT helper lives in `long_exposure/fanout.py` (a small deviation from the brief's suggested `exploration.py` inline placement, taken because `exploration.py` requires `prompt_toolkit` at import time and the test suite would not otherwise exercise the fix; functional behaviour is identical). Cycle-21 shadow ledgers under `/home/user/music-gen-instance/fork-392503ab7d47/clone-{0,1,2}/` were replayed through the new writer and reproduced the integrated main-ledger rows **byte-identically on full canonical JSON** — including `event_id`, which was source-swapped from random UUID4 to content-hashed UUID5 so the replay is now reproducible. `tests/test_harness_report_namespacing.py` (7/7 pass, one case above the brief's ≥ 6 asking price — the extra is an AST source-shape guardrail that prevents a future editor from re-inlining `uuid.uuid4()` or the raw f-string) and `tests/test_fanout_concat_validation.py` §16 (17/17 pass, extended by two cases — a regression cure and a regression guard). Existing suites remain green (writer 21/21, cross-branch integration §1-§30 0 failures, `promise_check` 0 ERRORS, `org_check` 0 ERRORS). Byte-determinism $\times 2$ on the replay concat holds at SHA-256 384326f7e17f.... The auditor's verdict is **VALIDATED**; the fan-out branch's scope is fully discharged.

Introduction

Cycle 21's post-merge integration for fork 392503ab7d47 diagnosed a concat-skip on Ledger-ConcatError: the harness auto-writes a per-clone `_run/report_cycles_<lo>-<hi>` row at reporting time, and across the three clones (0 → 1 → 2) the `ts` file-order was 16:53:17 → 16:59:57 → 16:54:07 — clone 2 finished its report before clone 1 did, violating per-candidate-milestone file-order monotonicity in the tightened concat. Cycle 22's worker patched the integration by serial `append_ledger_event` replay with per-clone id normalization (`_run/report_cycles_1-1_clone-{0,1,2}`), noted the root cause was harness behaviour rather than clone behaviour, and hoisted a durable upstream fix to a future cycle. This branch is that fix. The brief was clean: namespace the auto-write per clone at write time, land the change upstream under `long_exposure/*` (the established out-of-workspace WARN exemption), stay backwards-compatible with the root-cycle single-clone case, and prove the fix works by replaying the cycle-21 shadow ledgers through the new writer and reproducing the integrated main-ledger rows byte-identically at (mid, event_id, canonical_json-excl-ts).

Approach

Namespacing helper. `long_exposure/fanout.py` exports `report_cycles_milestone_id(lo, hi, env=None)` that returns `_run/report_cycles_<lo>-<hi>` on the root cycle (no fan-out context) and `_run/report_cycles_<lo>-<hi>_clone-<k>` when running under a `_CloneEnv(fork_id, k)` context derived from the `AGENT_FORK_ID` / `AGENT_FORK_CLONE_K` environment variables. The helper is a single call site; the upstream write-site (formerly an inline f-string on `_run/report_cycles_{lo}-{hi}`) now delegates to the helper. Backwards compatibility with the single-clone root-cycle case is preserved by construction — when `AGENT_FORK_CLONE_K` is unset, the helper returns the unqualified name and the root cycle keeps its historical row shape.

Event-id source-swap. Alongside the namespacing, the `event_id` for the auto-write is source-swapped from random UUID4 to content-hashed UUID5 via the `SSoT content_hash_event_id` helper. This makes the replay reproducible byte-identically (a random-UUID4 `event_id` would drift on every replay), and turns `append_ledger_event`'s duplicate guard into a semantic check (same payload → same `event_id` → duplicate correctly rejected) rather than a nominal check (fresh random ids would silently pass a duplicate write). Both are upside changes rather than compensations for the namespacing fix.

Backwards-compatible sentinel behaviour. `AGENT_FORK_CLONE_K=-1` (the sentinel the harness sometimes emits) produces `_run/report_cycles_<lo>-<hi>_clone--1` (double dash). This is cosmetic, matches the existing filename convention, and has no functional impact; the auditor flagged it as a pre-declared minor deviation.

Replay proof. `tests/test_harness_report_namespacing.py` case 4 reads the actual main ledger `promise_ledger.jsonl` and the actual cycle-21 shadow ledgers under `/home/user/music-gen-instance/fork-392503ab7d47/clone-{0,1,2}/`, applies the fix's transformation (namespace mid via `report_cycles_milestone_id` under a `_CloneEnv(fork_id="392503ab7d47", k=k)` context, drop the shadow row's `event_id`, re-derive via `content_hash_event_id`), and asserts byte-identity at (mid, event_id, canonical_json-excl-ts) AND ts. The `_replay_proof_cycle22.py` helper produced `mid=True eid=True full=True` for all three clones. This is a real end-to-end proof against the live cycle-22-integrated rows, not a sandboxed reconstruction.

Regression guards. Case 7 is an AST source-shape guardrail on the upstream write-site: the AST for the write-site must contain a call to `report_cycles_milestone_id` and must not contain either an inline `uuid.uuid4()` or the raw `f"_run/report_cycles_{...}"` f-string. Case 17 in the concat suite is the diagnostic-signal regression guard: if any future fanout sees `LedgerConcatError` on `_run/report_cycles_*` monotonicity, the failure signals the harness's env-var wiring

has regressed.

Findings

Test suites (all green under independent live re-run)

Suite	Result	Notes
tests/test_harness_report_namespacing.py (new)	7/7 PASS	Cases 1-7 including the AST source-shape guardrail on the upstream write-site and the full-canonical-JSON replay proof against the cycle-22-integrated rows.
tests/test_fanout_concat_validation.py §16 + §17	17/17 PASS	Case 16 = regression cure (namespaced auto-write concats cleanly with no per-clone normalization); case 17 = regression guard (future fanouts would fail loudly if the harness env-var wiring regressed). Writer regression clean.
tests/test_ledger_writer_validation.py	21/21 PASS unchanged	
tests/test_integration_cross_branch.py	0 failures across §1-§30	Final line result: PASS (0 failures).

Validators

- `promise_check` . — **0 ERROR** (WARN-only surface).
- `org_check` . — **0 ERROR** (pre-existing figures-in-docs WARNs only).

Cycle-21 shadow-ledger replay byte-identity

`_replay_proof_cycle22.py` reports `mid=True eid=True full=True` for clones 0/1/2 against the actual `promise_ledger.jsonl` rows and the actual cycle-21 shadow ledgers under `/home/user/music-gen-instance/fork-392503ab7d47/clone-{0,1,2}/`. The proof exceeds the brief's contract: **full canonical JSON identity** (`mid + eid + ts + every payload field`), not merely the nominated (`mid, event_id, canonical_json-excl-ts`) tuple.

Byte-determinism × 2 on the replay concat: SHA-256 384326f7e17f... equal on both fresh runs.

Deliverables on disk

- `docs/harness_report_namespacing_report.md` — 331 lines, all 10 required sections in order.
- `tests/test_harness_report_namespacing.py` — 7 cases (`_LE_PARENT` sys.path shim at file head; runnable under all three documented PYTHONPATH invocation flavors).
- `tests/test_fanout_concat_validation.py` — extended to 17 cases (case 16 regression cure + case 17 regression guard).
- `plan_of_record.md:107` — new 5-col row for `_infra/harness-auto-write-namespacing`.
- Upstream (out-of-workspace WARN exemption): `long_exposure/fanout.py` (new helper module); upstream write-site delegation to `report_cycles_milestone_id`.

Auditor’s verification-scope note on ledger events

The five ledger events the worker reports emitting (`_plan/register-...`, three `_infra/...`, and one `_archive/...`) do not appear in the workspace `promise_ledger.jsonl` yet — `grep` returns 0 matches. This is expected for a fan-out clone: events land in the clone’s shadow ledger under `/home/user/music-gen-instance/fork-cc548ca0c2e5/clone-0/` and are merged into the main ledger by the root conductor at post-merge integration. That path is outside this session’s read scope; however, the merge report exists at the expected path, and the ledger-event content-shape is exactly the SSoT-writer contract this branch just proved out. No basis to flag as a defect.

Minor deviations from the brief (all pre-declared, functionally neutral)

- **SSoT helper location.** Lives in `long_exposure/fanout.py` rather than being inlined in `exploration.py`; `exploration.py` requires `prompt_toolkit` at import time, so the test suite would not otherwise exercise the fix. Functional behaviour identical.
- **Test count.** 7 cases shipped (brief asked ≥ 6); the extra is the AST source-shape guardrail — pure defense-in-depth.
- **AGENT_FORK_CLONE_K=-1 sentinel** produces `_run/report_cycles_<lo>-<hi>_clone--1` (double dash). Cosmetic; matches existing filename convention; no functional impact.
- **WARN inflation to 115** (from baseline ≈ 17) is not from this branch — the workspace contains staged artefacts from concurrent siblings (`scripts/gen/batch_v4_*.py`, `scripts/ear/stability_*.py`, `tests/test_batch_v4_compound.py`, etc.); this branch’s own contribution is exactly one new orphan (`tests/test_harness_report_namespacing.py`) which will be adopted at merge when its ledger event lands.

Discussion

Three things about this branch are worth naming.

First, the branch retires the last known drift surface in the fan-out / concat pipeline. Cycle 10 hardened the writer gate; cycle 12 hardened the concat gate with per-milestone `ts` monotonicity and content-hash tiebreak; cycle 14 added `supersedes_path` type-check and the status enum; cycle 15 added state-transition validation via `_STATE_TRANSITIONS + validate_history`; and this branch closes the specific harness-behaviour that surfaced at cycle 21 and required a one-off reconciliation at cycle 22. Combined with `_infra/fanout-concat-hardening` (cycle 12) and `_infra/ledger-schema-hardening-v2` (cycle 14), the SSoT-writer chain (`writer` $\rightarrow$ `shadow` `lint` $\rightarrow$ `concat`) now enforces the same invariants at every boundary, and the specific `_run/report_cycles_*` collision cannot recur. The cycle-22 reconciliation driver at `tools/stale/_integrate_fork_392503ab7d47.py::replay_shadow` is now a historical artefact only; its three-line normalization pattern is no longer a template — future integrators should reach for the standard concat path directly.

Second, the `event_id` source-swap from random UUID4 to content-hashed UUID5 is worth calling out on its own. It was not the brief’s ask but it is a load-bearing upside: the replay is now reproducible byte-identically (a random UUID4 would drift on every replay and force the replay proof to compare partial tuples rather than full canonical JSON), and `append_ledger_event`’s duplicate guard now correctly rejects same-payload duplicates instead of accepting them under fresh random ids. This aligns the auto-write with the campaign’s broader determinism-by-content-hash discipline (`rule_id` in cycle 6, `event_id` in cycle 9, rule selection in cycle 10, salt tiebreak in cycle 11) and closes a small semantic hole in the deduplication story.

Third, the AST source-shape guardrail (case 7 in the new test module) is worth preserving as a pattern for other similar hazards. The fix is not a mechanical property of the runtime; a future editor could re-inline `uuid.uuid4()` or the raw f-string and the runtime tests would still pass because the behaviour would only regress under a fan-out clone. Case 7 asserts on the AST for the write-site — the call to `report_cycles_milestone_id` must be present, and neither an inline `uuid.uuid4()` nor the raw `f"_run/report_cycles_{...}"` f-string can be present. This is defense-in-depth against silent regression at edit time. Combined with the diagnostic-signal regression guard (case 17 in the concat suite), it means a future fanout that hits `LedgerConcatError` on `_run/report_cycles_*` monotonicity would be a clear signal that the harness env-var wiring (`AGENT_FORK_ID` / `AGENT_FORK_CLONE_K`) has regressed rather than a mysterious drift.

Nothing in this branch touches the uncalibrated CORN head or the rated-audio unblock, which remain the campaign's biggest open credibility gaps and are handled by the M-EAR-1 / M-INGEST-1 arm that awaits its `two-consecutive-media_ok=true` trigger.

Open Questions

Branch scope is genuinely exhausted. The following are legitimately future-cycle work, not this branch's:

- **Sentinel-value cosmetics.** `AGENT_FORK_CLONE_K=-1` producing `_run/report_cycles_<lo>-<hi>_clone--1` (double dash) is functionally neutral but arguably ugly. A one-liner change in `report_cycles_milestone_id` could special-case the sentinel; low priority.
- **Cycle-22 reconciliation driver retirement.** `tools/stale/_integrate_fork_392503ab7d47.py::replay_shadow` is now historical. A future cleanup cycle could delete it outright, but the auditor's guidance is that it can also stay in `tools/stale/` as a historical example — no urgency.
- **Extend the AST-guardrail pattern** to other single-call-site invariants the campaign relies on (SSoT event_id derivation call sites, `_lint_clone_shadow` invocation site inside `concat_clone_ledgers`, and so on) if similar silent-regression hazards emerge. Not urgent; deploy per hazard.
- **M-EAR-1 parent roll-up** and **CORN-head calibration** — still blocked on rated audio; will fire unattended through M-INGEST-1/egress-ready-automation when it triggers.

Appendix: Provenance

Cycle range: cycle 1 of fork `cc548ca0c2e5`, clone 0. **Working directory:** `/home/user/long-exposure-runs/music-gen`. **Session references:** researcher `48b1484b-8d71-4c4f-a6ac-0696ba53eaf4`, worker `7a74cb1c-bb52-4d8f-ba19-8f938975d8e6`, auditor `e757d28b-0a8a-4725-9724-529dab157ad6`. **Auditor verdict: VALIDATED.** Sub-milestone `_infra/harness-auto-write-namespacing` closes at `validated/high`.

Deliverables on disk.

- Upstream (out-of-workspace WARN exemption): `long_exposure/fanout.py` — new module exporting `report_cycles_milestone_id(lo, hi, env=None)` and `_CloneEnv(fork_id, k)`; upstream write-site refactored to delegate to the helper; `event_id` source-swapped to `content_hash_event_id`.
- Test: `tests/test_harness_report_namespacing.py` — 7 cases (root-cycle no-op, clone-context namespacing, sentinel behaviour, replay proof against cycle-21 shadows, byte-determinism $\times 2$, backwards-compatibility on the single-clone case, AST source-shape guardrail on the upstream write-site). `_LE_PARENT` sys.path shim at file head.
- Concat suite extension: `tests/test_fanout_concat_validation.py` §16 + §17 — regression cure and regression guard.
- Plan of record: `plan_of_record.md:107` — 5-col row for `_infra/harness-auto-write-`

namespacing.

- Report: docs/harness_report_namespacing_report.md (331 lines, 10 required sections).

Load-bearing runtime evidence.

- tests/test_harness_report_namespacing.py: 7/7 PASS live.
- tests/test_fanout_concat_validation.py: 17/17 PASS live (was 15; +2 cases).
- tests/test_ledger_writer_validation.py: 21/21 PASS unchanged.
- tests/test_integration_cross_branch.py: 0 failures across \$1-\$30.
- promise_check .: 0 ERRORS. org_check .: 0 ERRORS.
- Cycle-21 shadow-ledger replay: mid=True eid=True full=True for clones 0/1/2 against the actual promise_ledger.jsonl rows and actual cycle-21 shadow ledgers.
- Byte-determinism $\times 2$ on replay concat: SHA-256 384326f7e17f... equal on both fresh runs.
- Public API of append_ledger_event(workspace, event) and concat_clone_ledgers(workspace, fork_dir) $\rightarrow$ int byte-identical to cycle 15.
- SSoT is-identity preserved (_STATUS_ENUM is STATUS_VALUES; promise_check.REQUIRED_EVENT_FIELDS is _ledger_schema.REQUIRED_EVENT_FIELDS).

Ledger routing. Five shadow-ledger events emitted at /home/user/music-gen-instance/fork-cc548ca0c2e5/clone-0/promise_ledger.jsonl (_plan/register-harness-auto-write-namespacing, three _infra/*, _archive/harness-auto-write-namespacing-scratch). Clone-side orphan warnings will clear at post-merge integration via the standard concat_clone_ledgers path — no per-clone id normalization needed. Auditor's verification-scope note: shadow-ledger path is outside this session's read scope; the merge report exists at the expected path and the ledger-event content-shape matches the SSoT-writer contract this branch proved out.

Environment stack unchanged since cycle 10. mscore3 3.2.3 headless; Python 3.11.15; numpy 1.26.4; music21 9.1.0; mir_eval 0.8.2; fluidsynth (Debian) with pinned SF2 74594e8f...1cb0; DawDreamer + Surge XT Effects.vst3 at /usr/lib/vst3/; basic-pitch 0.4.0 in workspace/basic_pitch_venv/. Single-thread BLAS pins throughout.

Handoff. Merge report at /home/user/music-gen-instance/fork-cc548ca0c2e5/clone-0/merge_report.md. The root conductor's integration cycle should concat this clone's shadow ledger via the standard concat_clone_ledgers path (no per-clone id normalization needed — that is exactly what this branch retires). After merge, _infra/harness-auto-write-namespacing lands as validated/high and cycle-21 handoff #1 is retired. Case 17 stands as the permanent regression guard: if any future fan-out sees LedgerConcatError on _run/report_cycles_* monotonicity, the diagnostic signal is that the harness env-var wiring (AGENT_FORK_ID / AGENT_FORK_CLONE_K) has regressed.

validated